

Business Data

961731 Business Data Analytics

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06/07/2026

Agenda

- Introduction
- Types of Business Data
- Major Sources of Business Data
- Key Business Data Used in Data Analytics

Importance Of Data In Business

- Formulating Strategy



Bloomberg. (2020). "Amazon's Clever Machines Are Moving From the Warehouse to Headquarters".



The Economist. (2020). "Fashion forward: Zara's big idea".

Importance Of Data In Business

- Data in Operational Decision-Making



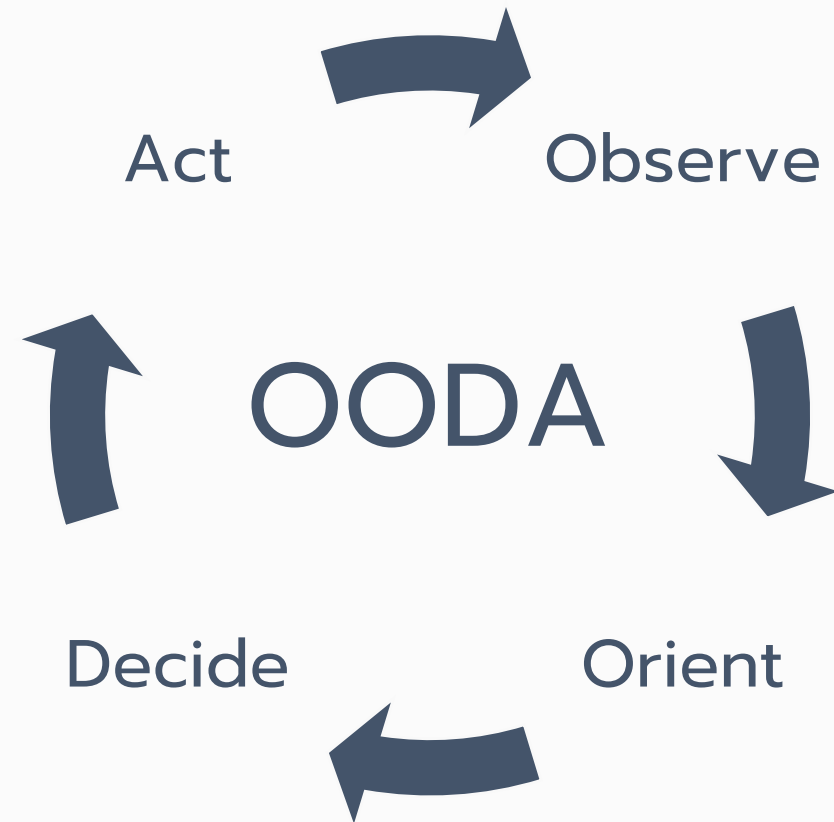
GE Reports. (2017). "The Brilliant Factory: Where Machines Talk To Each Other and People Tweet."



Toyota. (2020). "Toyota Production System."

Data-Driven Decision Making

- OODA Loop
 - Observe, orient, decide, and act
- Designed to handle uncertainty
- Invented by Colonel John Boyd



Data-Driven Decision Making

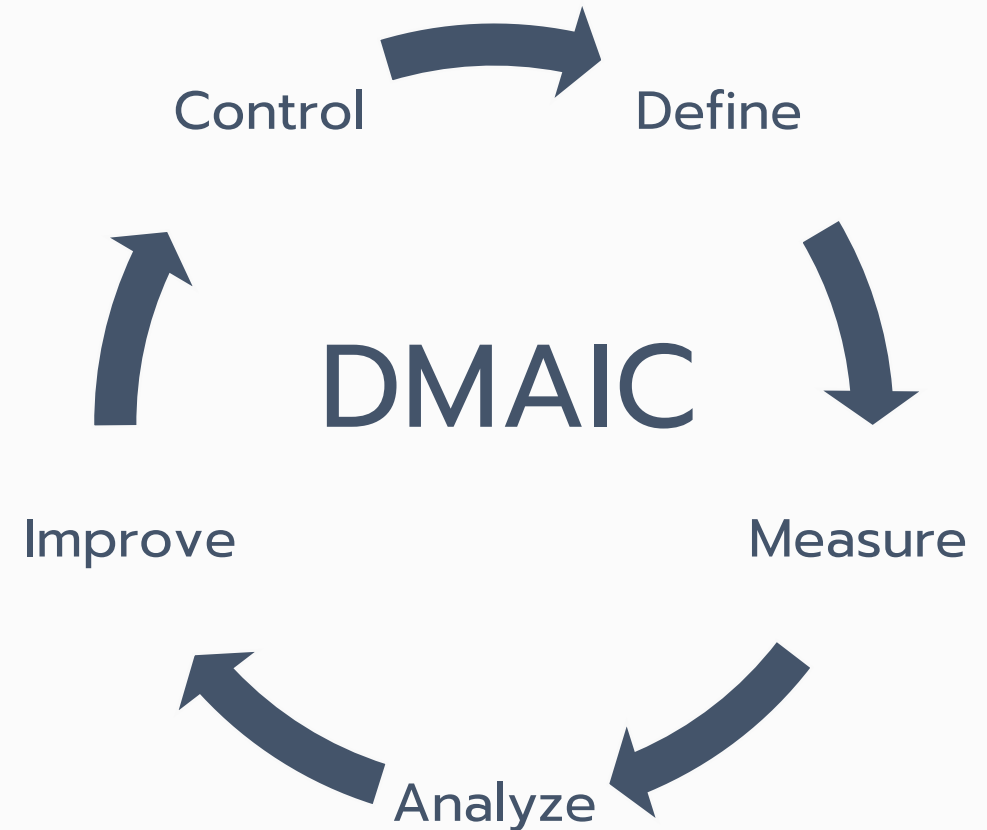
OODA Stage	Activities	Examples of Data
Observe	Collect and monitor data from various sources	Customer behavior data, market trends, financial data, operational data, feedback from employees
Orient	Analyze and interpret the data, identify opportunities or threats	Analysis results, patterns and trends in customer behavior, identified operational inefficiencies, financial performance insights
Decide	Evaluate options, consider potential outcomes, choose a course of action	Evaluation of different strategies, predictive models, decision trees, scenario analysis
Act	Implement the decision, execute the chosen action, observe the results	Outcomes of marketing campaigns, performance metrics after operational changes, financial results after implementing a new strategy

Data-Driven Decision Making

OODA Stage	Use Case 1: Amazon	Use Case 2: Toyota
Observe	Amazon collects customer behavior data, sales data, and other relevant market trends.	Toyota monitors machine performance, production rates, and quality data from the production line.
Orient	Amazon uses this data to understand customer preferences, market dynamics, and identify opportunities for product recommendations and new services.	Toyota interprets this data to identify potential issues, bottlenecks, or opportunities for process improvement in their production line.
Decide	Amazon decides on new features or changes to its platform, such as recommending products based on user behavior.	Toyota makes decisions to adjust machine settings, redesign a production process, or implement a new quality control measure.
Act	Amazon implements changes to its platform and observes user interaction and feedback.	Toyota implements the changes on the production line and observes the results, feedback from employees, and quality metrics.

Data-Driven Decision Making

- DMAIC
 - define, measure, analyze, improve and control
- Systematic approach to improvement
- Six Sigma



Data-Driven Decision Making

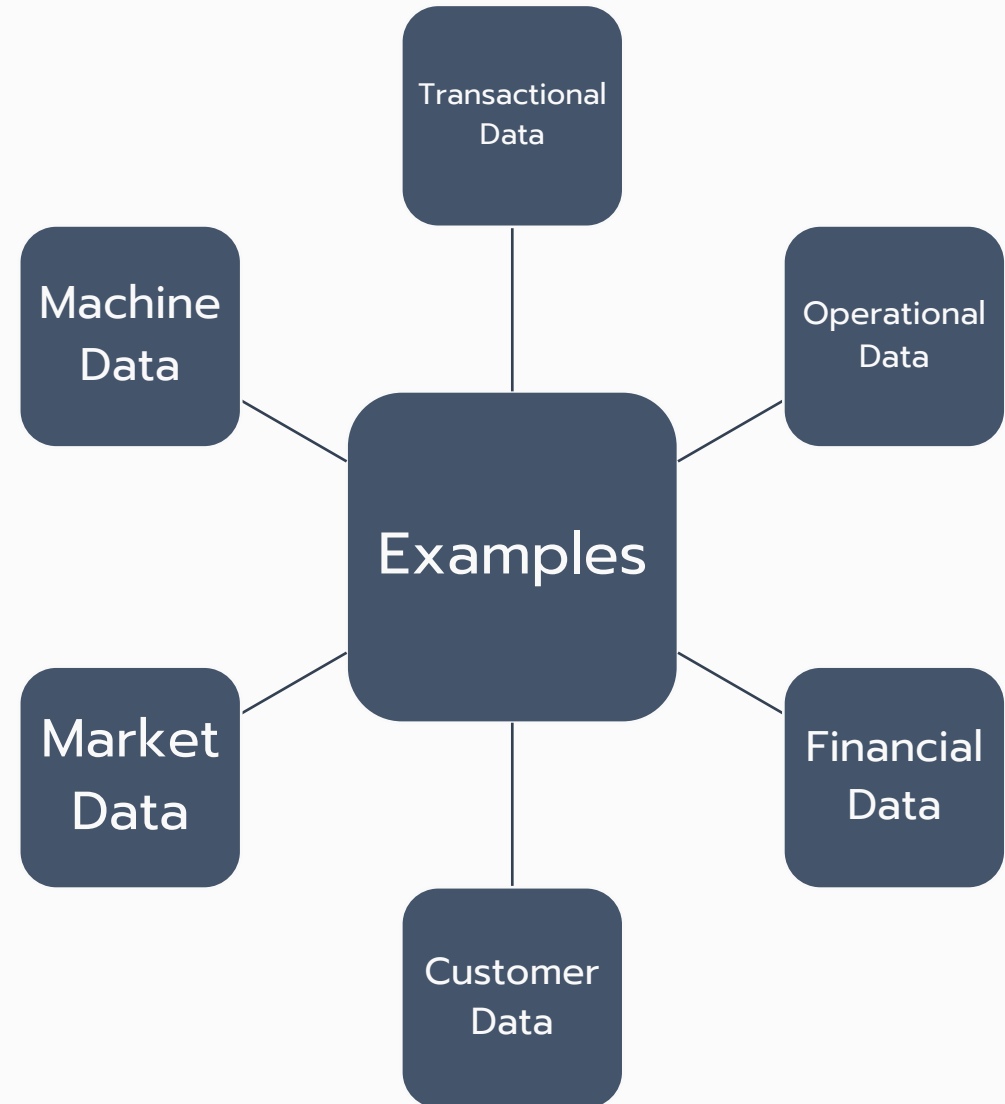
DMAIC Stage	Activities	Examples of Data
Define	Identify the problem, set objectives, and outline scope	Customer complaints, business goals, project boundaries
Measure	Collect relevant data, establish baseline performance	Operational performance data, customer satisfaction metrics
Analyze	Identify root causes of the problem, determine key factors	Process maps, cause-and-effect diagrams, correlation analysis results
Improve	Develop, test, and implement solutions	Results of solution testing, performance metrics after solution implementation
Control	Monitor the improved process, ensure the solution is maintained	Ongoing performance data, control charts, process documentation

Data-Driven Decision Making

DMAIC Stage	Use Case 1: General Electric	Use Case 2: Motorola
Define	General Electric (GE) identifies a problem with their manufacturing process which leads to increased production costs.	Motorola notices recurring defects in a particular model of their mobile devices.
Measure	GE measures the current performance and costs of the manufacturing process.	Motorola measures the number of defective units and identifies the common features of the defects.
Analyze	GE performs root cause analysis and identifies inefficiencies in their production line as the key problem.	Motorola analyzes the manufacturing process of the defective units and identifies a fault in the assembly line.
Improve	GE designs a streamlined production process, tests the new process, and implements it.	Motorola engineers design a solution to fix the assembly line fault, tests the solution, and implements it.
Control	GE monitors the improved process, observes a decrease in production costs, and standardizes the new process.	Motorola monitors the corrected assembly line, observes a decrease in defective units, and standardizes the improvements.

Types of Business Data

- Business data is collected, stored, and used by businesses to carry out various functions and processes.
- 2 Types of business data
 - Quantitative data
 - Qualitative data



Types of Business Data - Quantitative Data

- Numerical information that can be measured or counted.
- Quantitative data provides measurable, statistical insights to inform strategic and operational decisions.
- Examples of quantitative business data:
 - Financial Data: Revenue, expenses, profit margins, etc.
 - Operational Data: Production volumes, delivery times, etc.
 - Customer Data: Customer demographics, purchase frequency, etc.

Types of Business Data - Qualitative Data

- Non-numerical information that captures more subjective, descriptive insights.
- Qualitative data provides richer, more detailed insights to understand context, reasons, opinions, and motivations.
- Examples of qualitative business data:
 - Customer Data: Customer feedback, reviews, testimonials, etc.
 - Market Data: Industry trends, competitor strategies, social media sentiment, etc.
 - Operational Data: Employee feedback, observational data from production processes, etc.

Major Sources of Business Data

Source of Business Data	Description	Collection Method
Transactional Data	Data about sales, purchases, and other business transactions	Software platform such as POS systems, e-commerce platforms, and other transaction systems
Operational Data	Data about daily business operations, such as production volumes or delivery times	Process management software, supply chain management systems, or other operational technology
Financial Data	Financial information about revenue, expenses, profit margins, etc.	Financial management or accounting software, as well as through external financial reports and statements

Major Sources of Business Data

Source of Business Data	Description	Collection Method
Customer Data	Data about customers' demographics, behavior, and feedback	Customer relationship management (CRM) systems, surveys, social media, website analytics, and sales records
Market Data	Data about industry trends, competitor activity, and overall market conditions	Market research, industry reports, social media monitoring, and other external research tools
Machine Data	Data from industrial machinery and equipment, often used in manufacturing or logistics	IoT devices, industrial sensors, and other machine-to-machine technology

Reading assignment

Davenport, T.H., (2013). Analytics 3.0. Harvard Business Review.